Exp No. 1	Git Essentials – Commit, Push, Merge, and Pull Requests	REG. NO:URK23CS1307
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Objective
The aim is to demonstrate pushing changes to a GitHub repository, covering setup, managing visibility, and executing basic Git operations.

Tools/ Software Required
Docker desktop, Github, vs code,python

Description
1. Setting up Git: Ensure Git is installed on your computer.
2. Managing Repository Visibility: Understand the effects of making a repository private or public on
GitHub.
3. Changing Repository Visibility: Learn how to change the visibility of a repository on GitHub.
4. Pushing Changes to the Repository: Make changes to a local repository, commit them, and push them to the remote repository on GitHub.

Command list with explanation

Procedure
Introduction of GIT
Git is one of the ways of implementing the idea of version control. It is Distributed Version Control System.

Installing GIT
Before you start using Git, you have to make it available on your computer.it's already installed, it's probably a good idea to update to the latest version. You can either install it as a package or via another installer, or download the source code and compile it yourself.

Installing on Windows
There are also a few ways to install Git on Windows. The most official build is available for download on the Git website. Just go to HYPERLINK https://git-scm.com/download/win \hhttps://git-scm.com/download/win and the download will start automatically

To get an automated installation you can use the HYPERLINK https://community.chocolatey.org/packages/gitGit Chocolatey package.The easiest way to get Git is to download the executable from the Git website.Click 64-bit Git for Windows Setup to start the download, and then wait a moment — the download is only about 50 megabytes, so it shouldn take very long.Double-click the executable you just downloaded, then click Next to move through the installation prompts.

The first is the text editor Git will use. The default selection is Vim. Vim is ubiquitous and a hallmark of command-line interfaces everywh17ere but learning to use its idiosyncratic commands can be daunting. You should probably pick something else instead, like Visual Studio Code, Sublime, NotePad++, or any other plain text editor you like. The second is the way Git integrates itself into your PCs PATH. Make sure that the Git From The Command Line And Also From 3rd-Party Software is selected.

Managing Private and Public RepositoryMaking a repository Private

● GitHub will detach public forks of the public repository and put them into a new network. Public forks are not made private.

● If you using GitHub Free for personal accounts or organizations, some features wont be available in the repository after you change the visibility to private. Any published GitHub Pages site will be automatically unpublished. If you added a customdomain to the GitHub Pages site, you should remove or update your DNS records before making the repository private, to avoid the risk of a domain takeover.

● GitHub will no longer include the repository in the GitHub Archive Program.

● GitHub Advanced Security features, such as code scanning, will stop working.

Making a repository Public

● GitHub will detach private forks and turn them into a standalone private repository.

● If your converting your private repository to a public repository as part of a move toward creating an open source project.

● Once your repository is public, you can also view your repository community profile to see whether your project meets best practices for supporting contributors.

● The repository will automatically gain access to GitHub Advanced Security features.

Changing a repository's Visibility

1. On GitHub.com, navigate to the main page of the repository.
2. Under your repository name, click Settings. If you cannot see the settings tab,select the dropdown menu, then click Settings.
3. In the Danger Zone section, to the right of to Change repository Visibility , click Change visibility.
4. Select a visibility.
5. To verify that youre changing the correct repository visibility, type the name of the repository you want to change the visibility of.
6. Click I understand, change repository visibility.

Sample Inputs

```
Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git init
Reinitialized existing Git repository in C:/Users/Lenovo/myproject/.git/

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ touch text1.txt

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git add .

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   text1.txt


Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git log
commit d8efcb8e1bfc640d4098a9735c507d297130be2a (HEAD -> master)
Author: Chalice Garcia <felcigarcia@gmail.com>
Date:   Tue Feb 24 09:41:55 2026 +0530

    Initial commit

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git log --oneline
d8efcb8 (HEAD -> master) Initial commit

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (master)
$ git branch -M Main

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (Main)
$ git remote add origin https://github.com^C

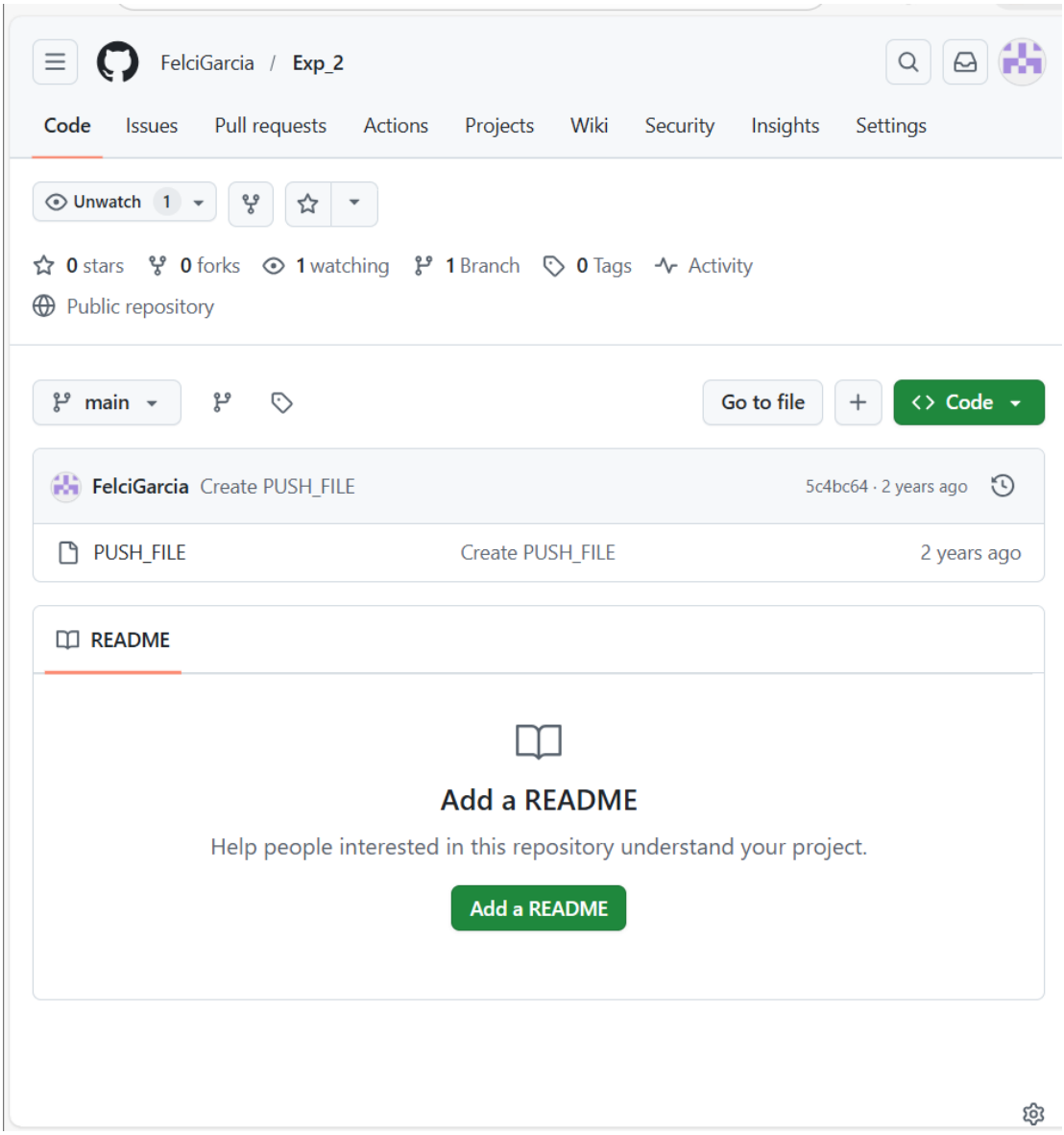
Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (Main)
$ git romote add origin https://github.com/FelciGarcia/Exp_2.git
git: 'romote' is not a git command. See 'git --help'.

The most similar command is
        remote

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (Main)
$ ^C

Lenovo@LAPTOP-6U4OOOCF MINGW64 ~/myproject (Main)
$ git remote add origin https://github.com/FelciGarcia/Exp_2.git
```

Expected Output / Observations



Result

Hence, Git Essentials Commit, Push, Merge, and Pull Requests are executed successfully